NATIONAL UNIVERSITY OF SINGAPORE

MA3236 - Nonlinear Programming

Final Examination

(Semester 1 : AY2024/2025)

Time allowed : 2 hours

INSTRUCTIONS TO CANDIDATES

1. Please write your matriculation/student number only. Do not write your name.
2. This examination paper contains **5** questions and comprises **4** printed pages.
3. Answer **ALL** questions.
4. Please start each question on a new page.
5. This is a CLOSED BOOK examination.
6. Students are allowed to use one double-sided, A4 size helpsheet.
7. Candidates may use scientific calculators. Programmable calculators are not allowed. However, they should lay out systematically the various steps in the calculations.

Question 1 [20 marks]

Consider the unconstrained minimization problem:

$$\min_{\mathbf{x}\in\mathbb{R}^2} f(\mathbf{x}) := \frac{1}{2}\mathbf{x}^T A\mathbf{x} + \mathbf{c}^T\mathbf{x},$$

where $A = \begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

(a) Given the initial point $\mathbf{x}^{(0)} = \begin{pmatrix} -1 \\ \frac{1}{2} \end{pmatrix}$, run one iteration of the steepest descent method with step size 1 to obtain $\mathbf{x}^{(1)}$.

(b) What happens if we use the steepest descent method with optimal step size in (a) instead? Determine the corresponding $\mathbf{x}^{(1)}$. Otherwise, explain why you can't get $\mathbf{x}^{(1)}$.

(c) Given the initial point $\mathbf{x}^{(0)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, run one iteration of the Newton method with step size 1 to obtain $\mathbf{x}^{(1)}$.

(d) Is the iterate $\mathbf{x}^{(1)}$ obtain in (b) a stationary point of f? If yes, determine if $\mathbf{x}^{(1)}$ is a local minimizer, local maximizer, or saddle point. Otherwise, justify why it is not a stationary point.

Solution.

(a)

$$x^{(0)} = \begin{pmatrix} -1 \\ \frac{1}{2} \end{pmatrix}, \quad \nabla f(x) = Ax + c = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix} \begin{pmatrix} -1 \\ \frac{1}{2} \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$x^{(1)} = x^{(0)} - \nabla f(x^{(0)}) = \begin{pmatrix} -1 \\ \frac{1}{2} \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 \\ 3/2 \end{pmatrix}$$

(b) $x^{(1)} = x^{(0)} - t\nabla f(x^{(0)}) = \begin{pmatrix} -1 - t \\ \frac{1}{2} + t \end{pmatrix}$

$$\begin{aligned} f(x^{(1)}) &= \tfrac{1}{2}(x^{(1)})^\top A x^{(1)} + c^\top x^{(1)} \\ &= \frac{1}{2}\begin{pmatrix} -1-t & \frac{1}{2}+t \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix}\begin{pmatrix} -1-t \\ \frac{1}{2}+t \end{pmatrix} - 1 - t \\ &= -\frac{1}{2}t^2 - 2t - \frac{5}{4} \end{aligned}$$

Hence, the function is not convex, and there is no minimizer.
We cannot find an appropriate $x^{(1)}$.

(c) $\mathbf{x}^{(0)} = \mathbf{0}$, $\nabla f(\mathbf{x}^{(0)}) = \mathbf{c} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $H_f(\mathbf{x}^{(0)}) = A$.

$$\mathbf{x}^{(1)} = \mathbf{x}^{(0)} - A^{-1}(A\mathbf{x}^{(0)} + \mathbf{c}) = -A^{-1}\mathbf{c} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

(d) It is a stationary point. It is a saddle point, because the Hessian is indefinite.

Question 2 [20 marks]

Consider the following constrained minimization problem in $\mathbb{R}^3$:

$$\begin{aligned} \min_{x_1,x_2,x_3} \quad & x_1 - x_1x_2 + x_3^2 \\ \text{s.t.} \quad & x_1 - x_2 \leqslant 1, \\ & x_1 + x_2 + x_3 = 2. \end{aligned}$$

(a) Determine the regularity at every feasible point.

(b) Find all the KKT points that satisfy the regularity condition (if any).

(c) For the KKT points found in (b), determine whether it is a local minimizer, local maximizer, or saddle point.

(d) Does the problem have a global minimizer? Justify your answer.

Solution.

(a) If $x_1 - x_2 < 1$, then $\nabla g(\mathbf{x}) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \neq \mathbf{0}$;

if $x_1 - x_2 = 1$, then $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \rho \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ yields a contradiction. Hence, every feasible point is regular.

(b) The KKT system is

$$\begin{cases} \begin{pmatrix} 1 - x_2 \\ -x_1 \\ 2x_3 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \mathbf{0}, \\ \mu \geqslant 0, \\ x_1 - x_2 \leqslant 1, \;\; x_1 + x_2 + x_3 = 2, \\ \mu(x_1 - x_2 - 1) = 0 \end{cases}$$

If $\mu = 0$, then

$$\begin{cases} x_1 = \lambda \\ x_2 = 1 + \lambda \\ x_3 = -\frac{\lambda}{2} \\ x_1 + x_2 + x_3 = 2 \end{cases} \Rightarrow \Big\{ x_1 = 2/3,\ x_2 = 5/3,\ x_3 = -1/3,\ \lambda = 2/3 \qquad \text{is a KKT point.}$$

If $\mu > 0$, then

$$\begin{cases} x_2 = x_1 - 1 \\ 1 - (x_1 - 1) + \mu + \lambda = 0 \\ -x_1 - \mu + \lambda = 0 \end{cases} \Rightarrow 2 + 2\mu = 0 \Rightarrow \mu = -1$$

This is not a valid KKT point.

(c)

$$H_L(\mathbf{x}) = H_f(\mathbf{x}) + \mu H_h(\mathbf{x}) + \lambda H_g(\mathbf{x}) = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

At the KKT point $(2/5, 5/3, -1/3)$, $h(\mathbf{x})$ is inactive. So we check the definiteness of $H_L(\mathbf{x})$ in the critical cone $C = \{\mathbf{y} | \nabla g(\mathbf{x})^T \mathbf{y} = 0\}$.

$$\nabla g(\mathbf{x}) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad Null(\nabla g(\mathbf{x})^T) = \{\mathbf{y} | y_1 + y_2 + y_3 = 0\} = \left\{ \begin{pmatrix} -s-t \\ s \\ t \end{pmatrix} : s, t \in \mathbb{R} \right\}$$

Take $Z(\mathbf{x}) = \begin{pmatrix} -1 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$, then

$$Z(\mathbf{x})^T H_L(\mathbf{x}) Z(\mathbf{x}) = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \text{ which is positive definite.}$$

So, the KKT point is a (strict) local minimizer.

(d) There is a global minimizer. Take $x_1 = 2 - x_1 - x_3$, then the optimization becomes

$$\begin{array}{ll} \min & 2 - 3x_2 - x_3 + x_2^2 + x_2 x_3 + x_3^2 \\ \text{s.t.} & -2x_2 - x_3 \leqslant -1 \end{array}$$

The objective function is coercive. On a closed set, there exists a global minimizer.

Question 3 [20 marks]

Given $c_{ij} \in \mathbb{R},\ a_i > 0,\ b_j > 0,\ i = 1, ..., n,\ j = 1, ..., n$ such that $\sum_{i=1}^n a_i = \sum_{j=1}^n b_j = 1$. Consider the following constrained minimization problem:

$$\begin{array}{ll} \min\limits_{x_{ij}} & \sum\limits_{i=1}^n \sum\limits_{j=1}^n c_{ij} x_{ij} \\ \text{s.t.} & \sum_{j=1}^n x_{ij} = a_i,\ i = 1, ..., n, \\ & \sum_{i=1}^n x_{ij} = b_j,\ j = 1, ..., n, \\ & X \in \mathbf{X} := \{X \in \mathbb{R}^{n \times n} \mid x_{ij} \geqslant 0,\ i = 1, ..., n,\ j = 1, ..., n\}. \end{array}$$

(a) Write down the Lagrangian function and compute the Lagrangian dual function.

(b) Write down the Lagrangian dual problem.

(c) Show that the Slater's condition holds true.

(d) Determine the duality gap.

Solution.

(a)

$$L(\mathbf{x}, \lambda, \hat{\lambda}) = \sum_{i=1}^{n}\sum_{j=1}^{n} c_{ij}x_{ij} + \sum_{i=1}^{n}\sum_{j=1}^{n} \lambda_i(x_{ij} - a_i) + \sum_{i=1}^{n}\sum_{j=1}^{n} \hat{\lambda}_j(x_{ij} - b_j)$$

So,

$$\begin{aligned}
\theta(\lambda, \hat{\lambda}) &= \inf_{x\in X} L(\mathbf{x}, \lambda, \hat{\lambda}) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} \inf_{x_{ij}\geqslant 0}\{(c_{ij} + \lambda_i + \hat{\lambda}_i)x_{ij}\} - \sum_{i=1}^{n} \lambda_i a_i - \sum_{j=1}^{n} \hat{\lambda}_j b_j \\
&= \begin{cases} -\sum_{i=1}^{n} \lambda_i a_i - \sum_{j=1}^{n} \hat{\lambda}_j b_j, & \text{if } c_{ij} + \lambda_i + \hat{\lambda}_j \geqslant 0,\ \forall\ i, j \\ -\infty, & \text{otherwise.} \end{cases}
\end{aligned}$$

(b)

$$\begin{array}{ll} \max & -\sum_{i=1}^{n} \lambda_i a_i - \sum_{j=1}^{n} \hat{\lambda}_j b_j \\ \text{s.t.} & \lambda_i,\ \hat{\lambda}_j \in \mathbb{R}\ \forall\ i, j, \\ & c_{ij} + \lambda_i + \hat{\lambda}_j \geqslant 0. \end{array}$$

(c) Take $x_{ij} = a_i b_j > 0$, then $\sum_{j=1}^{n} x_{ij} = a_i,\ \sum_{i=1}^{n} x_{ij} = b_j$. It means that $\mathbf{x}$ is strictly feasible. So, the Slater's condition holds true.

(d) The problem is convex and Slater's condition holds true. So, by strong duality theorem, there is 0 duality gap.

Question 4 [20 marks]

Consider the following convex constrained minimization problem:

$$\begin{array}{ll} \min & f(\mathbf{x}) := x_1^2 + x_2^2 - x_2 \\ \text{s.t.} & h_1(\mathbf{x}) := x_1 + x_2 - 1 \leqslant 0, \\ & h_2(\mathbf{x}) := -x_1 \leqslant 0, \\ & h_3(\mathbf{x}) := -x_2 \leqslant 0. \end{array}$$

We shall apply Frank-Wolfe algorithm to solve this problem:

(a) Starting from the initial point $\mathbf{x}^{(0)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, solve

$$\begin{array}{ll} \min & z(\mathbf{x}) = f(\mathbf{x}^{(0)}) + \nabla f(\mathbf{x}^{(0)})^T(\mathbf{x} - \mathbf{x}^{(0)}) \\ \text{s.t.} & h_i(\mathbf{x}) \leqslant 0,\ i = 1, 2, 3. \end{array}$$

(b) Define the search direction $\mathbf{d}^{(0)} = \hat{\mathbf{x}}^{(0)} - \mathbf{x}^{(0)}$, where $\hat{\mathbf{x}}^{(0)}$ is the solution of linear program in (a). Then, find the step size

$$t_0 = \arg\min_{0\leqslant t\leqslant 1}\ f(\mathbf{x}^{(0)} + t\mathbf{d}^{(0)}).$$

(c) Compute $\mathbf{x}^{(1)} = \mathbf{x}^{(0)} + t_0\mathbf{d}^{(0)}$.

(d) The Frank-Wolfe algorithm terminates when the gap between lower and upper bound is smaller than the given tolerance $\epsilon > 0$. The lower bound is determined by the optimal objective function value in (a), and the upper bound is determined by $f(\mathbf{x}^{(1)})$. Can we terminate the algorithm if $\epsilon = 1$? Justify your answer.

Solution.

(a) $\mathbf{x}^{(0)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\nabla f(\mathbf{x}) = \begin{pmatrix} 2x_1 \\ 2x_2 - 1 \end{pmatrix}$, $\nabla f(\mathbf{x}^{(0)}) = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$. Now, we solve the minimization problem

$$\begin{array}{rl} \min & 2x_1 - x_2 \\ \text{s.t.} & x_1 + x_2 - 1 \leqslant 0, \\ & x_1, x_2 \geqslant 0. \end{array}$$

The solution is $\hat{\mathbf{x}}^{(0)} = (0, 1)$, optimal value is -2.

(b) $\mathbf{d}^{(0)} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$, $\mathbf{x}^{(0)} + t\mathbf{d}^{(0)} = \begin{pmatrix} 1 - t \\ t \end{pmatrix}$. By minimizing $f(\mathbf{x}^{(0)} + t\mathbf{d}^{(0)})$, we have $t = 3/4$.

(c) $\mathbf{x}^{(1)} = \frac{1}{4}\begin{pmatrix} 1 \\ 3 \end{pmatrix}$.

(d) $f(\mathbf{x}^{(1)}) = -1/8,\ f(\mathbf{x}^{(1)}) - \hat{z} = -1/8 + 2 < 2$. Yes, it terminates.

Question 5 [**20 marks**] Consider the following problem:

$$\begin{array}{rll} \text{(P1)} & \min & f(\mathbf{x}) \\ & \text{s.t.} & h_i(\mathbf{x}) \leqslant 0,\ i = 1, ..., m. \end{array}$$

Suppose f, h_i, $i = 1, ..., m$ are differentiable and convex. Now consider a numerical method where we transform the constrained problem (P1) into an unconstrained problem by solving the auxiliary problem (P2):

$$\text{(P2)} \quad \min\ \phi(\mathbf{x}) := f(\mathbf{x}) + \alpha \max_{i=1,...,m} \max\{0, h_i(\mathbf{x})\},$$

where $\alpha > 0$ is a parameter.

(a) Argue that (P2) is equivalent to (P3):

$$\begin{array}{rll} \text{(P3)} & \min & f(\mathbf{x}) + \alpha y \\ & \text{s.t.} & h_i(\mathbf{x}) \leqslant y,\ i = 1, ..., m, \\ & & y \geqslant 0. \end{array}$$

(b) Write down the Lagrangian dual problem of (P3), and express it in terms of the Lagrangian dual function of (P1).

(c) Use the result in (b) to prove the following statement:

Suppose μ^* is an optimal solution of the Lagrangian dual of (P1), and that Slater's condition holds. If $\alpha > \mathbf{e}^T\mu^*$, then any solution of (P3) is also an optimal solution of (P2). Here, $\mathbf{e} = [1; 1; ...; 1]$ is a vector of ones.

(d) Although we can get an optimal solution of (P1) via solving (P2), it is not easy to tackle with (P2) in general because the objective function of (P2) might not be differentiable. Give an example where ϕ is not differentiable, but $f, h_i,\ i = 1, ..., m$ are differentiable.

Solution.

(a)

$$(P2) \equiv \min \quad f(\mathbf{x}) + \alpha y$$
$$\text{s.t.} \quad y = \max_{i=1,\ldots,m} \max\{0, h_i(\mathbf{x})\}$$

Suppose $(\mathbf{x}^*, y^*)$ is optimal to (P2), then optimal value is $f(\mathbf{x}^*) + \alpha y^*$.

y^* is the smallest value that satisfies $\begin{cases} y \geqslant 0 \\ y \geqslant h_1(\mathbf{x}) \\ \vdots \\ y \geqslant h_m(\mathbf{x}) \end{cases}$, which is equivalent to (P3).

(b) (P1): $\theta_1(\bar{\mu}) = \inf_{\mathbf{x}}\{f(\mathbf{x}) + \sum_i \bar{\mu}_i h_i(\mathbf{x})\}$

(P3):

$$\begin{aligned} \theta_3(\hat{\mu}) &= \inf_{\mathbf{x}, y \geqslant 0}\{f(\mathbf{x}) + \alpha y + \sum_i \hat{\mu}_i (h_i(\mathbf{x}) - y)\} \\ &= \theta_1(\hat{\mu}) + \inf_{y \geqslant 0}\{(\alpha - \sum_i \hat{\mu}_i) y\} \\ &= \begin{cases} \theta_1(\hat{\mu}), & \text{if } \alpha - \sum_i \hat{\mu}_i \geqslant 0 \\ -\infty, & \text{otherwise.} \end{cases} \end{aligned}$$

So, the dual problem is

$$(D) \quad \max \quad \theta_1(\mu)$$
$$\text{s.t.} \quad \alpha - \textstyle\sum_i \mu_i \geqslant 0,$$
$$\mu \geqslant 0$$

(c) The dual problem of (P1) is

$$(D1) \quad \max \quad \theta_1(\mu)$$
$$\text{s.t.} \quad \mu \geqslant 0$$

$(P1)$ is convex and the Slater condition's holds true. This implies that the strong duality holds for (P1). Since μ^* is an optimal dual solution for (D1), we can find an optimal solution $\mathbf{x}^*$ for (P1) such that

$$f(\mathbf{x}^*) = \theta(\mu^*).$$

Assuming $\alpha > \mathbf{e}^T \mu^*$, then

$$\begin{aligned} \theta_1(\mu^*) &\leqslant \text{optimal value of (D)} \quad \text{since } \mu^* \text{ is feasible for (D)} \\ &\leqslant \text{optimal value of (P3)} \quad \text{by weak duality} \\ &\leqslant f(\mathbf{x}^*) \quad \text{since } (\mathbf{x}^*, y = 0) \text{ is feasible for (P3)} \\ &= \theta_1(\mu^*) \quad \text{by strong duality on (P1).} \end{aligned}$$

This means we have equality throughout. By weak duality, μ^* is also an optimal solution to (D) and $(\mathbf{x}^*, 0)$ is an optimal solution to (P3).

Finally, by (a), (P2) is equivalent to (P3), so $\mathbf{x}^*$ is also a solution to (P2).

Remarks. The interpretation of this problem is that if α is large enough, then minimizing the penalized problem recovers an optimal solution of the constrained problem.

(d) The objective function is not differentiable in general. Any linear function could be an example.

End of Paper